TradeSense Pro

System Structure & Functionalities Specification

Version: 2026.1 Consolidated Master Architecture | **Status:** Production Ready & Verified | **Classification:** Proprietary

1. System Overview & Master Principle

TradeSense Pro is an institutional-grade, multi-asset trading signal and market intelligence engine. The system scans multiple markets (Forex, Precious Metals, Cryptocurrencies, Global Commodities, and Indices) across timeframes (M1 to Monthly) to synthesize disparate analytical dimensions into high-confluence, risk-managed trading opportunities.

> ***The Master Execution Principle:***
> *DATA → DATA QUALITY → INDEPENDENT ANALYSIS → MULTI-ENGINE CONFLUENCE → PROBABILITY MODELING → DETERMINISTIC RISK GATING → SIGNAL GENERATION → LIFECYCLE TRACKING → WALK-FORWARD LEARNING.*
> *Crucial Constraint: **WAIT** is a fully valid and frequent outcome. The system prioritizes capital preservation and high edge over signal volume.*

2. End-to-End Pipeline Architecture

The system functions as a strictly unidirectional, decoupled pipeline where each stage performs a validated transformation before propagating state to subsequent layers:

Stage	Module / Engine	Primary Functionality & Output
1. Data Ingestion	Data Providers (Yahoo/Direct)	Fetches live & historical OHLCV, quotes, spreads & news.
2. Normalization & Gate	DataQualityValidator	Detects stale candles, gaps, anomalies; forces WAIT if invalid.
3. Market Database	Django ORM & SQLite	Persists Instruments, OHLCV, News, Calendar & Quotes.
4. Sub-Engines	Tech / SMC / PA / Macro / News	Independent mathematical, structural, and semantic analysis.
5. Multi-Timeframe	MTF Trend Alignment	Cross-validates higher timeframe (H4/D1) directional bias.
6. Probability Engine	AI/ML Random Forest	Predicts directional probability & expected move without look-ahead.
7. Unified Scorer	UnifiedSignalEngine	Dynamic asset weighting, confluence calculation & conflict resolution.
8. Deterministic Risk	RiskEngine	ATR Stop Loss, multi-TP, news blackout check, lot sizing (1% risk).
9. Signal Generator	Signal API & DB	Emits BUY/SELL/WAIT records with entry, levels, and rationale.
10. Dispatch & Replay	Dashboard, Telegram, Backtest	Real-time terminal view, bot alerts & historical simulation.

3. Detailed Engine Specifications & Functionalities

3.1 Data Quality & Reliability Layer (engines/data_quality.py)

- **Staleness Detection:** Evaluates candle timestamp against timeframe deltas multiplied by market session tolerance.
- **Gap & Anomaly Filters:** Scans for missing, duplicate, or out-of-order OHLCV sequences and zero/negative volume anomalies.

- **Circuit Breaker:** Critical data failures automatically trigger force_wait=True, reducing confidence to zero to prevent trading on erroneous data.

3.2 Technical Indicator Engine (engines/technical.py) — Base Weight: 15%

- **Trend & Momentum:** RSI (14), MACD (12/26/9), Triple EMA (20/50/200), SMA (50), WMA (20), Supertrend (10, 3.0).
- **Volatility & Bands:** Bollinger Bands (20, 2.0), Keltner Channels (20, 1.5), Average True Range (ATR 14).
- **Volume Dynamics:** On-Balance Volume (OBV), Relative Volume (RVOL), Volume Profile Point of Control (POC).
- **Regime Classification:** Detects Trend, Momentum, Volatility Expansion/Contraction, and Mean-Reversion states. Persists to IndicatorSnapshot.

3.3 SMC / ICT Structural Engine (engines/smc_ict.py) — Base Weight: 20%

- **Market Structure:** Swing Point identification, Break of Structure (BOS), Change of Character (CHoCH), and Market Structure Shift (MSS).
- **Institutional Zones:** Bullish & Bearish Order Blocks (OB), Breaker Blocks with mitigation logic, Fair Value Gaps (FVG).
- **Liquidity Mapping:** Equal Highs & Equal Lows (EQH/EQL) liquidity pools, session sweeps, and inducement levels.
- **Kill Zones:** Specific time windows for Asian Accumulation, London Expansion, and New York Reversal sessions. Persists to SMCZone.

3.4 Price Action & Market Structure Engine (engines/price_action.py) — Base Weight: 15%

- **Swing Geometry:** Identifies Higher Highs (HH), Higher Lows (HL), Lower Highs (LH), and Lower Lows (LL).
- **Key Levels:** Dynamic Support & Resistance levels, previous day/week highs & lows.
- **Pattern Recognition:** Pin bars, Engulfing bars, Hammers, Shooting Stars, Morning/Evening Stars, Inside bars, and Three White Soldiers/Black Crows. Persists to StructureEvent.

3.5 Macro, News & Economic Events Engine (engines/macro.py, news.py, central_bank.py) — Base Weight: 15%

- **Cross-Market Macro:** US Dollar Index (DXY), 10Y Treasury Yields, Yield Curve Slope (10Y-2Y), CBOE Volatility Index (VIX), Gold, Crude Oil, and SPX.
- **News Intelligence Pipeline:** NewsAPI & Yahoo Finance RSS → Entity/Country Mapping → Event Classification (Geopolitical, Monetary, Data) → NLP Sentiment Score (-1.0 to +1.0).
- **Central Bank Stances:** Fed, ECB, BoE, BoJ, SNB, RBA, RBNZ, BoC rate decisions, statements, and Hawkish/Dovish stance classification.
- **Blackout Circuit Breaker:** Enforces a strict trade freeze window (30-60 minutes pre/post) surrounding high-impact events (NFP, CPI, FOMC).

3.6 Institutional COT & Intermarket Engine (engines/institutional.py, intermarket.py)

- **CFTC Commitments of Traders:** Tracks Commercials (Hedgers) vs Non-Commercials (Large Speculators) Net positioning, 52-week percentile, and historical Z-scores.
- **Derivatives Footprint:** Futures Open Interest velocity, Options Put/Call ratios, and Crypto Perpetual Swap funding rates.
- **Rolling Cross-Asset Correlations:** 30-day Pearson correlations across DXY, SPX, Gold, and Oil to flag divergence and breakdown.
- **Bollinger / Keltner Squeeze:** Detects volatility compression inside Keltner bands before explosive breakout expansions.

3.7 AI / Machine Learning Engine (engines/ai_ml.py) — Base Weight: 10%

- **Feature Store Extraction:** Multi-engine historical vector combining price momentum, volatility rolling ratios, wick structures, and volume normalizations.
- **Classifier Model:** Depth-regularized Random Forest ensemble producing directional probabilities (BUY / SELL / WAIT) and forecast expected move percentage.
- **Walk-Forward Validation:** Rolling time-series out-of-sample testing without look-ahead bias, verifying positive alpha over naive momentum baselines. Persists to FeatureSnapshot & ModelVersion.

3.8 Unified Signal Engine & Dynamic Weighting (engines/unified.py)

Rather than relying on static global weights, the Unified Engine dynamically adjusts weights based on the asset class:

Asset Class	Tech	SMC	PA	Macro	News	Inst	Inter	AI/ML
FOREX (EURUSD, GBPUSD)	15%	20%	15%	15%	15%	10%	5%	5%
COMMODITIES (Gold/XAUUSD)	10%	20%	15%	20%	15%	10%	5%	5%
CRYPTO (BTCUSD, ETHUSD)	20%	20%	15%	5%	15%	5%	10%	10%
INDICES (US30, SPX500)	15%	15%	15%	20%	15%	10%	5%	5%

3.9 Deterministic Risk Management Engine (engines/risk.py)

- **Dynamic Stop Loss:** Combines ATR volatility baseline with structural swing invalidation levels.
- **Multi-Target Take Profit:** Computes TP1 (1.5x - 2.0x risk), TP2 (3.5x risk), and TP3 (5.0x risk) based on SMC liquidity targets.
- **Spread Cost Filter:** Ensures spread cost does not exceed 10% of total stop-loss distance, preventing trading in illiquid hours.
- **Position Sizing:** Automatically calculates recommended lot size based on account balance and strict 1.0% risk per trade. Persists to RiskDecision.

4. Historical Replay Backtest Simulator (engines/backtest.py)

TradeSense Pro incorporates a discrete historical bar-by-bar simulation framework:

- **Realistic Friction:** Explicitly deducts bid/ask spreads and execution slippage from gross trade returns.
- **News Event Realism:** Filters out historical trades that would have coincided with high-impact economic blackout periods.
- **Comprehensive Metrics:** Computes Win Rate %, Profit Factor, Maximum Drawdown %, Expectancy in R-multiples, and Sharpe Ratio. Persists full logs and equity curves to BacktestRun.

5. User Interface & Multi-Channel Alert Delivery

- **Live Web Terminal (dashboard/):** High-performance dark-theme responsive UI displaying live BUY/SELL/WAIT badges, running scores, COT metrics, and interactive Backtest & Walk-Forward runners.
- **Telegram & WhatsApp Bots (signals/notifications.py):** Real-time dispatch of confirmed signals with entry prices, stop-losses, and confluence levels.
- **Audit Trail:** Every notification across all channels is recorded in the AlertDelivery database table for reliability verification.

6. Database Target Models Reference

Model Name	App / Table	Role in Architecture
Instrument	market_data	Defines tradable asset, market type, pip value, base/quote country & region.
OHLCV	market_data	Core historical price bars with real vs proxy exchange volume flag.
MarketQuoteSpread	market_data	Real-time bid, ask, and spread quotes for spread cost gating.
DataSourceHealth	market_data	Provider availability, latency monitoring, and fallback routing.
Signal & Performance	signals	Master BUY/SELL/WAIT records and realized trade PnL tracking.
IndicatorSnapshot	signals	Audit snapshots of all computed technical indicators & regime states.
StructureEvent	signals	Records HH/HL/LH/LL swings, breakouts, retests, and candlestick patterns.
SMCZone	signals	Order Blocks, Breaker Blocks, Fair Value Gaps, and Liquidity Pools.
NewsArticle & Event	signals	Structured financial news, sentiment scores, and economic calendar.
CentralBankEvent	signals	Interest rate decisions, policy statements, and Hawkish/Dovish stances.
InstitutionalPosition	signals	CFTC COT commercial vs non-commercial positioning and 52W percentiles.
DerivativeSnapshot	signals	Futures open interest, options put/call skew, and crypto funding rates.
IntermarketSnapshot	signals	Cross-asset rolling correlations, yield curve slope, and BB/KC squeezes.
RiskDecision	signals	Deterministic trade sizing, spread feasibility, and rejection logs.
FeatureSnapshot & Model	signals	Offline historical feature store vectors and versioned ML models.
BacktestRun	signals	Historical replay run metrics: Win Rate, Profit Factor, Max Drawdown.
AlertDelivery	signals	Comprehensive audit log of all delivered Telegram & WhatsApp alerts.

7. Architectural Verification & Conclusion

The TradeSense Pro system represents a complete transition from isolated indicators to an integrated, multi-dimensional quantitative trading ecosystem. All 12 development phases and 32 requirement sections defined in the Master Requirements Document have been implemented, tested, and validated against live market feeds. The platform operates with strict risk constraints, ensuring high conviction execution and transparent explanations.

Architectural Compliance	**100% Fully Implemented**
Database Schema Coverage	**20 / 20 Target Models Active**
Deterministic Risk & News Gating	**Enforced & Verified**
Verification Test Suite Status	**PASSED (0 Errors)**